

Bypassing ratio and its measurement in macropore soils

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ABSTRACT

Bypassing ratio is defined as a ratio of the rate of macropore flow to the application rate which must be higher than the infiltration rate into the soil matrix. Four methods were used to measure the bypassing ratio in four soils in the Danubian Lowland: clay soil in Jurová, loamy soil in Macov and Most near Bratislava, and loamy-sand soil in Kalinkovo. It was found out that the bypassing ratio is high both at the soil surface and below the root zone during the whole vegetation season. Taking into account the extensive measurements made by Šútor (1986), the mean infiltration rate into the soil matrix was set to be 2 mm.hr⁻¹ for this region. As it was calculated by Faško (Lichner et al. 1999) from 10-year-lasting observations, the rain with intensity higher than 2 mm.hr⁻¹ and duration more than 1 hr can appear 24-times in the average in the south-western Slovakia during vegetation season. That rain in the combination with a high value of bypassing ratio can be the cause of a drop of fertilization efficiency, as well as the shallow groundwater pollution in the Danubian Lowland.

Keywords: macropore flow; bypassing ratio; Danubian Lowland

Water applied to the soil surface at rates higher than the infiltration rate into the soil matrix could pass rapidly down the surface-vented macropores and disappear to the sub-soil without bringing a benefit to roots of plants, and without wetting and leaching the soil between macropores. The above-mentioned rapid macropore flow is the cause of a drop of irrigation and fertilization efficiency, as well as an increase of the shallow groundwater vulnerability to agrochemicals. At the same time, the matrix flow of water occurs in the soil at the rates up to four orders of magnitude slower than the macropore flow (Beven and Germann 1982). A dichotomy of the real soil is a basis of two-domain models of water flow and solute transport (Jarvis 1994). Neglecting the effects of macropore flow can be the cause of underestimation of the position of infiltration and redistribution front with the models based on the Richards' equation for a homogeneous soil (Štekauerová et al. 1999).

The bypassing ratio *BR* is defined as a ratio of the rate at which water bypasses the soil matrix (i.e. the rate of macropore flow) to the application rate (rain or irrigation intensity) (Kneale and White 1984). It is a useful characteristic of the rapid vertical movement of water and solutes along macropores through an unsaturated soil matrix. As it is a relative value, it can be used for a comparison of the effects of macropore flow in various soils, depths, etc., and therefore, as an indicator of a drop of irrigation and fertilization efficiency, as well as the shallow groundwater pollution (Lichner et al. 1999).

In this paper, four methods were used to measure the bypassing ratio and its variation in various soils under various crops during vegetation season. Combination of those results with the probability of heavy rain (with the intensity higher than the mean infiltration rate into the soil matrix and duration more than 1 hr) was used to make conclusions about the shallow groundwater vulnerability from agrochemicals in the Danubian Lowland.

MATERIAL AND METHODS

The Danubian Lowland is a large (1260 km²) agriculturally utilized area, with shallow (0.5–3 m) underlying aquifer containing about 10 km³ of freshwater. In this region, the bypassing ratio and its temporal variations were assessed on light, medium heavy, and heavy soil in the course of vegetation season. The light soil occurs in Kalinkovo [loamy-sand soil, Eutric Fluvisol (FAO 1994)]. The medium heavy soil occurs in Macov and Most near Bratislava [loamy soil, Calcari-mollic Fluvisol (FAO 1994)]. The heavy soil occurs in Jurová [clay soil, Calcari-mollic Fluvisol (FAO 1994)]. Basic characteristics of the soils under study were estimated by Fulajtár et al. (1998).

Four methods were used to measure the bypassing ratio and its variation in various soils under various crops during vegetation season. As the first method, the laboratory method for bypassing ratio measurement designed by Kneale and White (1984) was used. In this method, presumption of no outflow from the soil matrix serves to make the rate of outflow equal to the rate of macropore flow.

As the second method, the way similar to that designed by Watson and Luxmoore (1986) was used. The hydraulic conductivity of the soil matrix $K(-3 \text{ cm})$ at the tension $h = -3 \text{ cm}$ was measured with the tension infiltrometer, the saturated hydraulic conductivity of the soil K_s was measured with the double-ring infiltrometer (inner ring of 30-cm diameter and buffer ring of 55-cm diameter), and the bypassing ratio was calculated by:

$$BR = [K_s - K(-3 \text{ cm})]/K_s \quad (1)$$

As the third method, the way similar to that designed by Wopereis et al. (1994) was used. In their design, shallow surface tillage (0–50 mm) was used to reduce the

macropore flow in a cracked clay soil. In our design, the double-ring infiltrometer (inner ring of 30-cm diameter and buffer ring of 55-cm diameter) was used to measure both the steady flow through an undisturbed soil surface [equal to the saturated hydraulic conductivity K_s (Šútor 1986)], and, after manual destroying the structural pores near the soil surface in the inner ring, the steady flow through the textural pores only (equal to the hydraulic conductivity K_{sd}). Bypassing ratio BR_d was calculated by:

$$BR_d = (K_s - K_{sd})/K_s \quad (2)$$

The relative error δ_{BR} of the BR_d measurement was calculated by:

$$\delta_{BR} = (BR_d - BR)/BR \quad (3)$$

where the value BR was deemed to be the true value.

In the fourth method, the leading edge of radioactive tracer ^{131}I was formed below the area of maximum plant uptake, and its displacement h was measured with the

Geiger-Müller detector. The bypassing ratio was calculated by:

$$BR = 1 - hn_{ef}/I \quad (4)$$

where: I is the cumulative infiltration of water resulting in the displacement h of the radioactive tracer leading edge formed in the soil matrix, $n_{ef} = \Theta_s - \Theta_r$ is the effective porosity of the soil under study, Θ_s is the saturated water content, and Θ_r is the residual water content of the soil matrix (Lichner 1997, 1998).

RESULTS AND DISCUSSION

The bypassing ratio measurement with the first method was undertaken in the undisturbed loamy soil column (20-cm diameter and approximately 35 cm long) sampled in the untilled field covered with grass in Most near Bratislava in September 2000. Water was applied at the soil surface with the rain simulator composed of the water applicator with 66 needles and a peristaltic pump. Outflow from the column began 33 minutes after starting the irrigation (with an intensity of 2.8 cm.hr⁻¹), but it did not reach a steady rate as it was observed in case of clay soil columns (Kneale and White 1984; Van Stiphout et al. 1987; Wopereis et al. 1994). As a result, the bypassing ratio increased from 4–90% in the course of 2-hr lasting measurement. It was concluded that the laboratory method for bypassing ratio measurement designed by Kneale and White (1984) fails in the case of medium heavy and light soils where the presumption of no outflow from the soil matrix cannot be fulfilled.

In the vegetation season 2000, the bypassing ratio measurements were undertaken in the soils under winter wheat with the second and third measuring method. From the results taken with the second method it was found out that the bypassing ratio BR varied in the topsoil between 89.2–99.8% in Jurová, 53.6–98.2% in Macov, and 76.8–98.2% in Kalinkovo (Tables 1–3 and Figure 1). The highest variations of BR were in the loamy soil in Macov. The values of BR were very high in the clay soil in Jurová during the whole vegetation season due to enormously high temperatures and low precipitation resulting in the formation of desiccation cracks. The values $K(-3 \text{ cm}) > K_s$, as measured in Kalinkovo on May 18 and September 20, 2000, cannot be explained by the authors. An impact of the harvest (decrease of the BR -values in all the three soils) and shallow ploughing (increase of the BR -values in all the three soils) were in contrary to our anticipation.

As it can be seen in Tables 1–3, relative error δ_{BR} of the BR_d measurements was high in loamy and loamy-sand soil. It should be noted that closing the ports of surface-vented macropores was most effective in the soil with a high clay content (almost 60% of particles $\leq 0.01 \text{ mm}$ was found in the surface layer of the soil in Jurová, Table 4).

In 1993–1996 the bypassing ratio was measured in the loamy soil in Most near Bratislava with the radioactive tracer technique [together with the TDR-measurements

Table 1. Hydraulic conductivity, bypassing ratio and relative error δ_{BR} as measured in the clay soil in Jurová

Date	3. 5. 2000	25. 5. 2000	11. 7. 2000	16. 8. 2000	19. 9. 2000
$K(-3 \text{ cm}) (\text{m.s}^{-1})$	$9.03 \cdot 10^{-6}$	$1.47 \cdot 10^{-6}$	$1.83 \cdot 10^{-6}$	$3.17 \cdot 10^{-6}$	$1.42 \cdot 10^{-6}$
$K_s (\text{m.s}^{-1})$	$8.33 \cdot 10^{-5}$	$6.67 \cdot 10^{-4}$	$1.67 \cdot 10^{-4}$	$2.22 \cdot 10^{-4}$	$1.67 \cdot 10^{-4}$
$K_{sd} (\text{m.s}^{-1})$	$3.13 \cdot 10^{-6}$	$3.71 \cdot 10^{-5}$	$2.32 \cdot 10^{-5}$	$1.67 \cdot 10^{-5}$	$1.23 \cdot 10^{-5}$
$BR (\%)$	89.2	99.8	98.9	98.6	99.1
$BR_d (\%)$	96.2	94.4	86	92.5	92.6
$\delta_{BR} (\%)$	7.8	-5.4	-9.1	-6.2	-6.6

Table 2. Hydraulic conductivity, bypassing ratio and relative error δ_{BR} as measured in the loamy soil in Macov

Date	3. 5. 2000	25. 5. 2000	11. 7. 2000	16. 8. 2000	19. 9. 2000
$K(-3 \text{ cm}) (\text{m.s}^{-1})$	$1.08 \cdot 10^{-5}$	$5.83 \cdot 10^{-7}$	$1.19 \cdot 10^{-6}$	$3.22 \cdot 10^{-6}$	$1.50 \cdot 10^{-6}$
$K_s (\text{m.s}^{-1})$	$2.33 \cdot 10^{-5}$	$1.67 \cdot 10^{-5}$	$7.83 \cdot 10^{-6}$	$3.33 \cdot 10^{-5}$	$8.33 \cdot 10^{-5}$
$K_{sd} (\text{m.s}^{-1})$	$2.78 \cdot 10^{-6}$	$3.92 \cdot 10^{-6}$	$2.78 \cdot 10^{-6}$	$1.15 \cdot 10^{-5}$	$2.22 \cdot 10^{-5}$
$BR (\%)$	53.6	96.5	84.8	90.3	98.2
$BR_d (\%)$	88.1	76.5	64.5	65.5	73.3
$\delta_{BR} (\%)$	39.2	-20.7	-23.9	-27.5	-25.4

Table 3. Hydraulic conductivity, bypassing ratio and relative error δ_{BR} as measured in the loamy-sand soil in Kalinkovo

Date	18. 5. 2000	6. 6. 2000	13. 7. 2000	15. 8. 2000	20. 9. 2000
$K(-3 \text{ cm}) (\text{m.s}^{-1})$	$8.92 \cdot 10^{-6}$	$5.83 \cdot 10^{-7}$	$1.16 \cdot 10^{-5}$	$1.14 \cdot 10^{-6}$	$1.94 \cdot 10^{-5}$
$K_s (\text{m.s}^{-1})$	$8.33 \cdot 10^{-6}$	$3.33 \cdot 10^{-5}$	$5.00 \cdot 10^{-5}$	$2.33 \cdot 10^{-5}$	$4.17 \cdot 10^{-6}$
$K_{sd} (\text{m.s}^{-1})$	$2.78 \cdot 10^{-6}$	$5.13 \cdot 10^{-6}$	$1.01 \cdot 10^{-5}$	$9.01 \cdot 10^{-6}$	$2.78 \cdot 10^{-6}$
$BR (\%)$	—	98.2	76.8	95.1	—
$BR_d (\%)$	66.6	84.6	79.8	61.4	33.3
$\delta_{BR} (\%)$	—	-13.8	3.9	-35.4	—

of macropore flow in 1995 (Alaoui et al. 1997)]. It was found out that the macropore flow was about 49% of the total flow ($BR = 0.49$) as measured in the depth of $z = 48 \text{ cm}$ in a spring barley field in April–May 1993, $BR = 0.15$ in the depth of $z = 56 \text{ cm}$ in a maize field in July 1995, $BR = 0.55$ in the depth of $z = 78 \text{ cm}$ in the grassland soil in August 1996. It can be seen that the warm and rainless second half of April 1993 (similar situation repeated in April 2000) resulted in the crack formation and as high macropore flow in the barley field as in the untilled soil with well-developed macropore systems.

The bypassing ratio measurement was repeated in the same untilled field covered with grass in September 2000 using the third method. From the measurements of the steady flow through an undisturbed soil surface and the steady flow through the textural pores only in the depth

of $z = 40 \text{ cm}$ with the double-ring infiltrometer techniques, $BR_d = 0.31$ was estimated.

The surface-vented macropores can lead water and solutes rapidly to the subsoil if the rain or irrigation intensity is higher than the infiltration rate into the soil matrix. Taking into account the extensive measurements made by Šútor (1986) in the Danubian Lowland, the mean infiltration rate into the soil matrix was set to be 2 mm.hr⁻¹ for this region. As it was calculated by Faško (Lichner et al. 1999) from 10-year-lasting observations, the rain with intensity higher than 2 mm.hr⁻¹ and duration more than 1 hr can appear 24-times in the average in the south-western Slovakia during vegetation season (April–October). That rain can be the cause of rapid transport of surface-applied agrochemicals down the macropores and shallow groundwater pollution in the Danubian Lowland, and

Table 4. Textural composition of the surface layer (0.1–0.2 m) of soils in Jurová, Macov, and Kalinkovo (Fulajtár et al. 1998)

Site	Sand (%)			Silt (%)		Clay (%)
	$\geq 0.25 \text{ mm}$	0.25–0.05 mm	0.05–0.01 mm	0.01–0.001 mm	$\leq 0.001 \text{ mm}$	$\leq 0.01 \text{ mm}$
Jurová	1.54	11.35	27.84	37.32	21.92	59.27
Macov	0.86	36.14	28.84	19.94	14.22	34.16
Kalinkovo	6.04	55.77	22.48	10.22	5.49	15.71

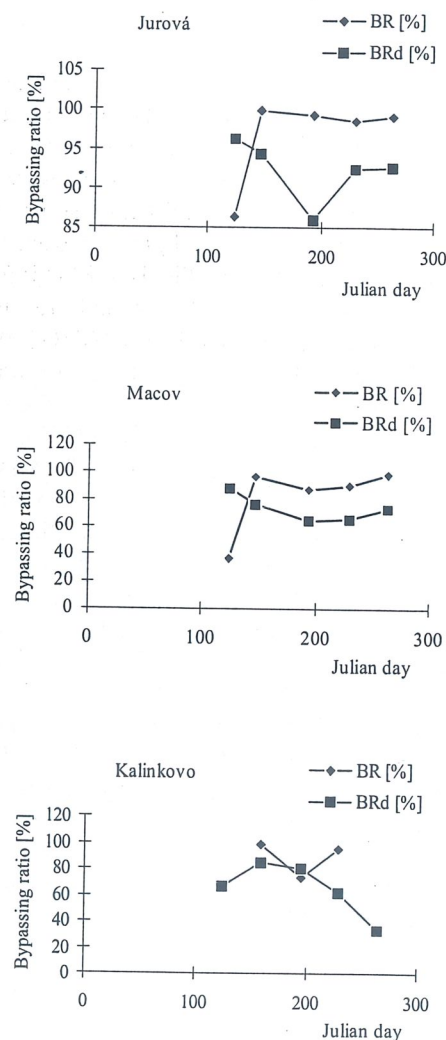


Figure 1. Variations of the bypassing ratio as measured in three soils in the vegetation season 2000; the values of BR and BR_d were calculated using eqs. (1) and (2), respectively

therefore, the subsurface fertilizer banding [e.g. with the fertilizer applicator designed by Ressler et al. (1997)] should be used in this region.

CONCLUSIONS

Bypassing ratio yields an information about the part of rain or irrigation water which could pass rapidly down the surface-vented macropores and disappear to the subsoil without being of benefit to roots of plants, and without wetting and leaching the soil between macropores. It can serve as an indicator of a drop of irrigation and fertilization efficiency, as well as the shallow groundwater pollution.

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ABSTRAKT

Měření obtokového podílu v půdě

Obtokový podíl je definovaný jako podíl rychlosti proudění v makropórech a rychlosti infiltrace vody do půdy. Použili jsme čtyři metody měření obtokového podílu ve čtyřech půdách v Podunajské nížině: v jílovité půdě v Jurové, v hlinité půdě v Macově a v Mostu u Bratislavy a v hlinitopísčité půdě v Kalinkově. Zjistili jsme, že obtokový podíl byl vysoký jak na povrchu půdy, tak i v podkořenové oblasti během celé vegetační sezony. Z rozsáhlých měření, která prováděl Šútor (1986), jsme stanovili střední hodnotu infiltrační rychlosti do půdní matrice pro tuto oblast, která je 2 mm.h^{-1} . Faško (Lichner et al. 1999) z desetiletých pozorování srážek zjistil, že déšť s intenzitou vyšší než 2 mm.h^{-1} a trváním delším než 1 h se v oblasti jihozápadního Slovenska vyskytne během vegetační sezony průměrně 24krát. Tento déšť ve spojení s vysokou hodnotou obtokového podílu může způsobit pokles účinnosti hnojení a kontaminaci mělkých podzemních vod agrochemikáliemi.

Klíčová slova: proudění v makropórech; obtokový podíl; Podunajská nížina

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